Mu Runtime Reference

version *0.2.19*

type keywords and aliases

supertype	*T*	
bool	(),:nil are false, otherwise true	
condition	*keyword,* see **exceptions**	
list	:cons or (),:nil	
ns	#s(:ns #(:t *fixnum symbol*))	
ns-designator	*ns,* :nil, :unqual	
:null	(),:nil	
:char	*char*	8 bit ASCII
:cons	*cons, list*	list, cons, dotted pair
:fixnum	*fixnum, fix*	56 bit signed integer
:float	*float, fl*	32 bit IEEE float
:func	*function, fn*	function
:keyword	*keyword, key*	symbol
:stream	*stream*	file or string type
:struct	*struct*	see ***structs***
:symbol	*symbol, sym*	LISP-1 symbol
:vector	*vector, string, str*	typed vector
		:bit :char :t
		:byte :fixnum :float

core

apply *fn list*	*T*	apply *fn* to *list*
compile *form*	*T*	*mu* form compiler
eq *T T'*	*bool*	*T* and *T'* identical?
eval *form*	*T*	evaluate *form*
type-of *T*	*key*	type keyword
view *for*	*vector*	vector of object
fix *fn T*	*T*	fixpoint of *fn*
gc	*bool*	garbage collection
repr *T*	*vector*	tag representation
unrepr *vector*	*T*	tag representation

tag *vector* is an 8 element :byte vector of little-endian argument tag bits.

special forms

:lambda *list . list' function*	anonymous *fn*	
:alambda *list . list'*	*function*	anonymous *fn*
:quote *T*	*list*	quoted form
:if *T T' T''*	*T*	*conditional*

frames

frame binding: (*fn* . #(:t ...))

%frame-stack	*list*	active *frames*
%frame-pop *fn*	*frame*	pop *function's* top frame binding
%frame-push *frame*	*cons*	push frame
%frame-ref *fn fix*	*T*	*function*, offset

symbols

boundp *sym*	*bool*	is *symbol* bound?
make-symbol *string*	*sym*	uninterned *symbol*
symbol-namespace *sym*	*ns-designator*	*namespace designator*
symbol-name *symbol*	*string*	*name binding*
symbol-value *symbol*	*T*	value binding

fixnums

add *fix fix'*	*fixnum*	sum
ash *fix fix'*	*fixnum*	arithmetic shift
div *fix fix'*	*fixnum*	quotient
less-than *fix fix'*	*bool*	*fix < fix'?*
logand *fix fix'*	*fixnum*	bitwise and
lognot *fix*	*fixnum*	bitwise complement
logor *fix fix'*	*fixnum*	bitwise or
mul *fix fix'*	*fixnum*	product
sub *fix fix'*	*fixnum*	difference

floats

fadd *fl fl'*	*float*	sum
fdiv *fl fl'*	*float*	quotient
fless-than *fl fl'*	*bool*	*fl < fl'?*
fmul *fl fl'*	*float*	product
fsub *fl fl'*	*float*	difference

conses/lists

append *list*	*list*	append lists
car *list*	*T*	head of *list*
cdr *list*	*T*	tail of *list*
cons *T T'*	*cons*	(*T* . *T'*)
length *list*	*fixnum*	length of *list*
nth *fix list*	*T*	*n*th *car* of *list*
nthcdr *fix list*	*T*	*n*th *cdr* of *list* \

vectors

make-vector *key list*	*vector*	specialized *vector* from list
vector-length *vector*	*fixnum*	length of *vector*
vector-type *vector*	*key*	type of *vector*
svref *vector fix*	*T*	*n*th element

namespaces

mu *(static)*, ***keyword*** *#s(:ns #(:t str))*

make-namespace *str*	*ns*	make *namespace*
namespace-name *ns*	*string*	*namespace* name
intern *ns str value*	*symbol*	intern symbol in non-static *namespace*
find-namespace *str*	*ns*	map *string* to *namespace*
find *ns string*	*symbol*	map *string* to *symbol*

structs

make-struct *key list*	*struct*	type *key* from *list*
struct-type *struct*	*key*	*struct* type *key*
struct-vec *struct*	*vector*	of *struct* members

streams

standard-input	*stream*	std input *stream*
standard-output	*stream*	std out *stream*
error-output	*stream*	std error *stream*
open *type dir str bool*	*stream*	open *stream,* raise error if *bool*

type	:file :string
dir	:input :output :bidir

close *stream*	*bool*	close *stream*
openp *stream*	*bool*	is *stream* open?
flush *stream*	*bool*	flush *steam*
get-string *stream*	*string*	from *string stream*
read-byte *stream bool T*	*byte*	read *byte* from *stream,* error on eof, *T:* eof-value
read-char *stream bool T*	*char*	read *char* from *stream,* error on eof, *T:* eof-value
unread-char *char stream*	*char*	push *char* onto *stream*
write-byte *byte stream*	*byte*	write *byte*
write-char *char stream*	*char*	write *char*
read *stream bool T*	*T*	read *stream*
write T *bool stream*	*T*	write with escape

exceptions

with-exception *fn fn'*	*T*	catch exception

fn - (:lambda (*obj cond src*) **.** *body*)
fn' - (:lambda () **.** *body*)

raise *T T'*	*keyword* raise exception on *T* from source designator *T'* with *keyword* condition

:arity :div0 :eof :error :except
:future :ns :open :over :quasi
:range :read :exit :signal :stream
:syntax :syscall :type :unbound :under
:write :storage :user

Features

[features]
default = ["core", "env", "system"]

feature/core	***core***	*list*	core state
	sleep	*fixnum*	microseonds
	process-mem-virt	*fixnum*	vmem
	process-mem-res	*fixnum*	reserve
	process-time	*fixnum*	microseconds
	time-units-per-sec	*fixnum*	
	namespace *string*	*list*	*symbol* list
feature/env	***heap-info***	*()*	heap info to stdout
	heap-room *key*	*vector*	allocations
	#(:t size total free ...)		
	heap-size *key*	*fixnum*	type size
	load *string*	*bool*	load source
	cache-room	*vector*	allocations
	#(:t size total ...)		
feature/system	***uname***	:t	system info
	shell *string list*	*fixnum*	shell command
	exit *fixnum*		doesn't return
	sysinfo	:t	not on macOS
feature/instrument			
	instrument-control *key*		:on|:off|:get
	key | vec		

environment config

JSON config format:

```
{
  "pages": N,
  "gc-mode": "none" | "auto",
}
```

Mu library API

[dependencies]
mu = {
git = "https://github.com/Software-Knife-and-Tool/mu.git",
branch = "main"
}

```
use mu::{ Mu, Env, Config, Result, Tag, Exception, Condition };

impl Mu {
    fn compile(_: &Env, _: Tag) → Result<Tag>
    fn config(_: Option<String>) → Config
    fn env(_: &Config) → Env
    fn eq(_: Tag, _: Tag) → bool;
    fn err_out() → Tag
    fn eval_str(_: &Env, _: &str) → Result<Tag>
    fn eval(_: &Env, _: Tag) → Result<Tag>
    fn exception_string(_: &Env, _: Exception) → String
    fn load(_: &Env, _: &str) → Result<bool>
    fn env(_: &Config) → Env
    fn nil() → Tag
    fn read_str(_: &Env, _: &str) → Result<Tag>
    fn read(_: &Env, _: Tag, _: bool, _: Tag) → Result<Tag>
    fn std_in() → Tag
    fn std_out() → Tag
    fn version() → &str
    fn write_str(_: &Env, _: &str, _: Tag) → Result<()>
    fn write_to_string(_: &Env, _: Tag, _: bool) → String
    fn write(_: &Env, _: Tag, _: bool, _: Tag) → Result<()>
}
```

API function argument details:

compile &Env ***T*** → Result<Tag>
config Option<String> → Config
env &Config → Env
eq T T' → bool
err_out → Tag
eval_str &Env &str → Result<Tag>
eval &Env ***T*** → Result<Tag>
exception_string &Env Exception → String
load &Env &str → Result<bool>
env &Config → Env
nil → Tag
read_str &Env &str → Result<Tag>
read &Env ***stream*** bool ***T*** → Result<Tag>
// bool – raise exception on end of stream
std_in → Tag
std_out → Tag
version → &str
write_str &Env &str ***stream*** → Result<()>
write_to_string &Env *T* bool→ String
// bool – print escaped
write &Env *T* bool ***stream*** → Result<()>
// bool – print escaped

Reader Syntax

;	comment to end of line
#|...|#	block comment
'*form*	quoted form
`*form*	backquoted form
`(...)	backquoted list (proper lists)
,*form*	eval backquoted form
,@*form*	eval-splice backquoted form
(...)	constant *list*
()	empty *list*, prints as :nil
(... . .)	dotted *list*
"..."	*string, char* vector
\	single escape in strings
ns:name	qualified *symbol,* where *ns* and *name* are *symbol constituents*
name	lexical *symbol*
#*	bit vector
#x...	hexadecimal *fixnum*
#.*form*	read-time eval
#\...	*char*
#(:type ...)	*vector*
#s(:type ...)	*struct*
#:...	uninterned *symbol*
"`,;"	terminating macro char
#	non-terminating macro char
!$%&*+-.	*symbol constituent*
<>=?@[]|	
:^_{}~/	
A..Za..z	
0..9	
	character designators
0x09 #\tab	
0x0a #\linefeed	
0x0c #\page	
0x0d #\return	
0x20 #\space	